Vaibhavi Shambhu Shetty

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EDUCATION

North Carolina State University, Raleigh, NC — August 2023 - May 2025

Master of Computer Science — **GPA 3.8/4**

Coursework: Software Engineering, Automated Learning and Data Analysis, Design and Analysis of Algorithms, Object Oriented Design and Development, Parallel Systems, Neural Networks and Deep Learning

Vishwakarma Institute of Technology, Pune, India — August 2019 - May 2023

Bachelor of Technology, Information Technology (Honors in Artificial Intelligence and Data Analytics) — **CGPA 9.68/10**

Coursework: Database and Management Systems, Cloud Computing, Operating Systems, Machine Learning

TECHNICAL SKILLS

- **Languages:** Python, Typescript, JavaScript, C/C++, Java, HTML, CSS, ReactJS, SQL, PHP, Ruby, SuiteScripts
- **Frameworks and Systems:** TensorFlow, scikit-learn, NumPy, Flutter, NetSuite, MongoDB, AWS, MySQL, Ruby on Rails
- **Other:** CI/CD, Android and IOS Development, Firebase, Devops, Docker, Kubernetes, Git

WORK EXPERIENCE

Graduate Research Assistant, North Carolina State University, US — August 2024 - Present

- NC State **Sustainability** funded Campus as A Classroom Program, eSCoOp researcher.
- Developed scalable data pipelines using **Python** (Pandas, NumPy) for microbial abundance analysis in compost samples.
- Engineered interactive visualizations with **Matplotlib**, **Seaborn**, and **Plotly**.
- Analyzed microbial growth curves using **R** (**growthcurver, Metacoder**) to interpret compost sample growth dynamics.

Junior Associate - Technology Intern, Synechron, Sunrise, FL, US — May 2024 - August 2024

- Developed a robust anomaly detection system leveraging **Isolation Forest** for real-time process intelligence, handling complex datasets with over 1 million records.
- Engineered a comprehensive data validation pipeline using **Python**, ensuring compliance with schema requirements, leading to a 30% improvement in data preprocessing efficiency.
- Integrated **AWS S3** for scalable data retrieval, optimizing data handling and reducing local storage dependencies by 50%.
- Integrated SHAP for model explainability and MLflow for model versioning and experiment tracking.

Technical Intern Co-op, **Appficiency,** Pune, India — January 2023 - June 2023

- Designed and customized Oracle **NetSuite Suitecloud** systems for different client requests using **SuiteScripts**
- Improved the **SharePoint** architecture at Provision Point by 80% by identifying security vulnerabilities

ACADEMIC PROJECTS

Course Manager (Python, MySQL, Typescript)

- Designed a comprehensive ER diagram and implemented a robust **MySQL** database schema with stored procedures, triggers, and functions for data integrity and automation interactive learning management system for seamless online course creation.
- Developed a dynamic, user-friendly interface using **React** and **Redux** with **TypeScript**, ensuring smooth state management and responsive user experience and engineered the backend with **Flask**.

Open-Source project: Expertiza (Typescript, Ruby)

- Developed type-safe, role-based Institution and Notification Management **UIs** using **TypeScript** and **ReactJS** and implemented a robust backend in **Ruby on Rails** with object-oriented principles, securing role-specific access.
- Built a real-time notification system with unread alerts, toggles, and tailored student views, leveraging **design patterns** in Ruby and validating functionality through comprehensive **RSpec** tests.

Cloud Architecture of Online Bookstore Application

- Designed and implemented a robust **microservices-based cloud architecture solution** for an online bookstore application model, leveraging **AWS** services to incorporate efficient load balancing and auto-scaling capabilities.
- Executed the **Horizontal Pod Auto-scaling (HPA)** feature on a **MiniKube** cluster, dynamically fine-tuning capacity by responding to CPU utilization. Performed experiments within the **MiniKube** environment to validate load balancing, utilizing the Locust tool for generating and analyzing traffic patterns.

EXTRACURRICULAR

Active member of **Women in Computer Science** at North Carolina State University — August 2023 - Present

Research Paper published in Scopus Indexed Journal – PETI, Taiwan. Link